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Protocol

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An inexpensive semi-automated sample processing pipeline for cell-free RNA extraction

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Abstract

Despite advances in automated liquid handling and microfluidics, preparing samples for RNA sequencing at scale generally requires expensive equipment, which is beyond the reach of many academic laboratories. Manual sample preparation remains a slow, expensive and error-prone process. Here, we describe a low-cost, semi-automated pipeline to extract cell-free RNA using one of two commercially available, inexpensive and open-source robotic systems: the Opentrons OT1.0 or OT2.0. Like many RNA isolation protocols, ours can be decomposed into three subparts: RNA extraction, DNA digestion and RNA cleaning and concentration. RT–qPCR data using a synthetic spike-in confirms comparable RNA quality to the gold standard, manual sample processing. The semi-automated pipeline also shows improvement in sample throughput (+12×), time spent (−11×), cost (−3×) and biohazardous waste produced (−4×) compared with its manual counterpart. This protocol enables cell-free RNA extraction from 96 samples simultaneously in 4.5 h; in practice, this dramatically improves the time to results, as we recently demonstrated. Importantly, any laboratory already has most of the parts required (manual pipette and corresponding tips and kits for RNA isolation, cleaning and concentration) to build a semi-automated sample processing pipeline of their own and would only need to purchase or three-dimensionally print a few extra parts (US$5.5 K–12 K in total). This pipeline is also generalizable for many nucleic acid extraction applications, thereby increasing the scale of studies, which can be performed in small research laboratories.

Key points

- A protocol for setting up, calibrating and validating a semi-automated sample processing pipeline for cell-free RNA isolation from clinical samples using the Opentrons 1.0 or 2.0 open-source robotic platform.
- The key benefits of this semi-automated system, as compared with manual extraction, include an increase in sample throughput by 12 fold. When compared with other available protocols for automated RNA extraction, this protocol provides a complementary option for larger sample volumes (>0.5 mL).

Key references

Moufarrej, M. N. et al. *Nature* **602**, 689–694 (2022): https://doi.org/10.1038/s41586-022-04410-z

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Introduction

Liquid biopsies offer an unprecedented opportunity to study human disease on a molecular scale and sample from the entire human body at once[1–6]. Applications of this technique span vast fields such as oncology[7–9], transplant care[10,11] and prenatal diagnostics[12–16]. Numerous groups and companies have now demonstrated the utility of 'omic'-based blood analysis in proof-of-concept work[4,8,11,17–19]. However, scaling such discovery research remains obstructed by the lack of robust sample processing, which is paramount to yield trustworthy results—especially for human samples that may come from sources with disparate collection and storage guidelines.

Most discovery-focused work has four main stages: sample collection, sample processing, data generation and data analysis. Sample collection, data generation and data analysis have progressed tremendously due to the advent of mature biobanks, massively parallel sequencing (e.g., NovaSeq) and cloud-based computing (e.g., Amazon Web Services, Google Cloud), respectively. Sample processing, however, has remained manual despite increasing sample numbers. As a result, this laborious step constrains the number of samples processed at once and can lead to batch effects.

Although the biotechnology industry has long moved to robotic liquid handling solutions[20], many laboratories have not. Most automated solutions for laboratory work available today focus on production quality pipelines and do not accommodate research-based laboratory work in at least one of these ways: (1) no high-volume pipetting (e.g., Bravo), (2) no easy user-level customization (e.g., Hamilton, Eppendorf) and (3) high initial expense (US$20,000–100,000) and custom consumables (e.g., Bravo, Hamilton, Eppendorf).

In this protocol, we describe the design and validation of a semi-automated pipeline to extract cell-free RNA (cfRNA) by outfitting an existing inexpensive, open-source robotic platform, the Opentrons 1.0 (OT1), or using the Opentrons 2.0 (OT2). Specifically, we detail how to setup and calibrate the robotic system, validate the platform end-to-end using an exogenous RNA control and, finally, to use the system to process clinical samples in high throughput with suitable quality for downstream analysis[14]. We also show that RT–qPCR data on control RNA samples purified using this platform are comparable to results obtained with the gold standard used in prior work—manual sample processing[13]. Such robust and scalable solutions will allow us to test hypotheses faster and spend less time on repetitive but important tasks such as sample processing.

Experimental design

Robotic system overview

Like many RNA isolation protocols, processing cfRNA samples can be decomposed into three subparts: RNA extraction, DNA digestion, and RNA cleaning and concentration (Fig. 1). Our process requires a single robotic system than can easily fit within a chemical hood—necessary to process bodily fluids—and can be configured to use any individual eight-channel pipette. Nearly all steps are performed on the robot with minimal user input. At the system's prompting, the user fills reagent boats, transfers plates for centrifugation or swaps out tip boxes. Sample addition is performed by hand to easily accommodate various sample storage strategies (e.g., 1.5 mL microcentrifuge tubes, matrix tubes). Similarly, DNAse master mix and eluant addition are also performed by hand to maintain the system's simplicity since these steps are the only ones in the process that require small-volume (<15 µL) pipetting. By using standard laboratory equipment and interfacing with the user at key steps, our semi-automated pipeline streamlines sample processing while maintaining the flexibility that is key to discovery-stage research.

Robotic system setup

OT1. The OT1 system is composed of motors that drive motion in the *x*, *y* and *z* directions across ten deck positions where the user can place reagent boats, plates, tip boxes and trash (Fig. 2a,d). To use the robot, the user must mount their pipette of choice so that it is rigidly fixed in all directions and only moves with the motors themselves. We achieved this by designing and three-dimensional (3D) printing several parts to work with our pipette of choice.

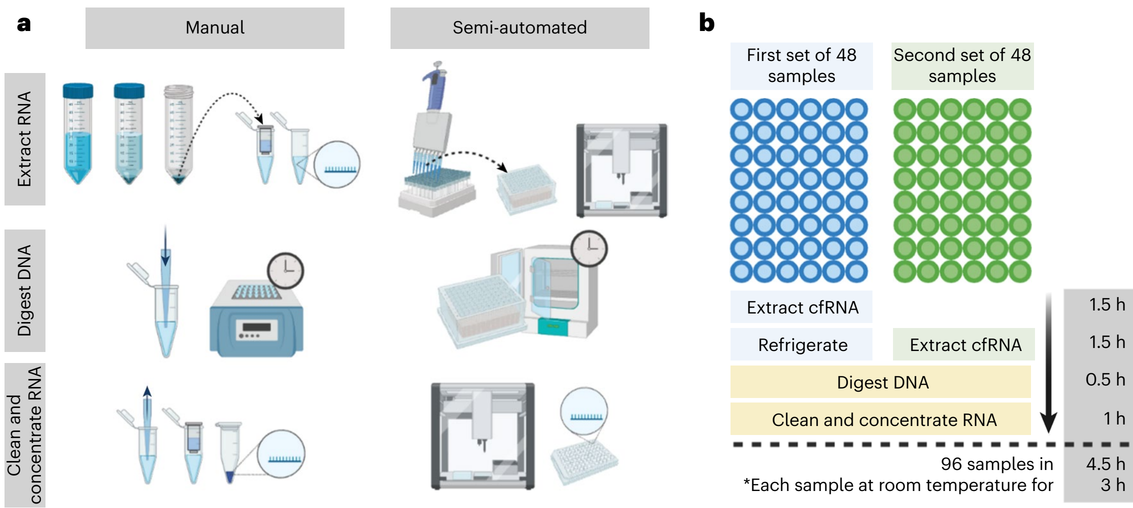

Fig. 1 | Comparison of manual and semi-automated cfRNA extraction protocol. a, Extracting cfRNA from 1 mL of plasma requires 5 mL of added reagents, yielding a maximum total volume of 6 mL. In the manual protocol, this calls for the use of 50 mL conical tubes; typically, eight samples are processed at once. In the semi-automated protocol, to maintain system scalability, we opted to use 48-deep-well plates. Processing 96 samples using 48-well plates requires that the samples are split into two batches briefly and then recombined into one plate before DNAse digestion. Samples are then cleaned and concentrated either in individual 1.5 mL tubes (manual) or a 96-well plate (semi-automated). **b**, Overall protocol flow and time estimates for each stage of the protocol. Blue and green circles denote separate 48-well plates. Figure created with BioRender.com.

To fix motion in the *x* direction, we mounted two U-shaped clasps (Fig. 2b, dark blue arrows). These clasps are then fastened to a hard back (Fig. 2b, light blue box). This back board coupled with a piece mounted on top of the pipette (Fig. 2b, light blue arrow) prevents unexpected *z*-axis motion during tip retrieval and disposal. Finally, two more L-shaped holds (Fig. 2b, green arrows) prevent *y*-axis motion in the pipette body, which can otherwise rotate freely. Together, these 3D-printed fixtures allow the system to remain properly calibrated across multiple runs while mounting and disposing of tips to aspirate, mix, transfer and dispense reagents (Fig. 2c, full protocol at https://youtu.be/g6RsSaNvSNA). Additionally, we designed and printed a holder for pipette tip boxes to fit within a standard deck footprint. For more details regarding pipette installation, see https://support.opentrons.com/en/articles/689945ot-one-installing-pipettes. The OT1 system is no longer commercially available. We provide guidance on how to build a similar system from scratch in the 'Equipment setup'.

OT2. Like the OT1, the OT2 is composed of motors that drive motion in the *x*, *y* and *z* directions. Unlike the OT1, the OT2 has 11 deck positions and a fixed trash position (Fig. 2d) and comes with several ready-to-use pipettes including an eight-channel 300 µL pipette (200 µL max volume with filter tips). Consequently, the same 1 mL pipetting step for OT1 will be split up into multiple serial pipetting steps in the OT2-compatible version.

Robotic system calibration

OT1. Once the pipette has been fixed onto the system, it is necessary to calibrate the OT1, and register the *x*, *y* and *z* position for each consumable in each deck position it may occupy (Fig. 2e). To this end, we have provided a series of calibration scripts and instructions at https://github.com/miramou/cfRNA_pipeline_automation/tree/master/Opentrons that can be used with the OT1 software to calibrate the system. Special care should be taken when calibrating the pipette tip boxes as even small miscalibrations may result in a failure to pick up tips or faulty tip placement with negative downstream consequences such as mispipetting. More details can be found at https://support.opentrons.com/en/articles/689977ot-one-calibrating-the-deck.

OT2. Like the OT1, you will have to calibrate the OT2 and register the *x*, *y* and *z* positions for each consumable in each deck position (Fig. 2e). More detailed instructions can be found at

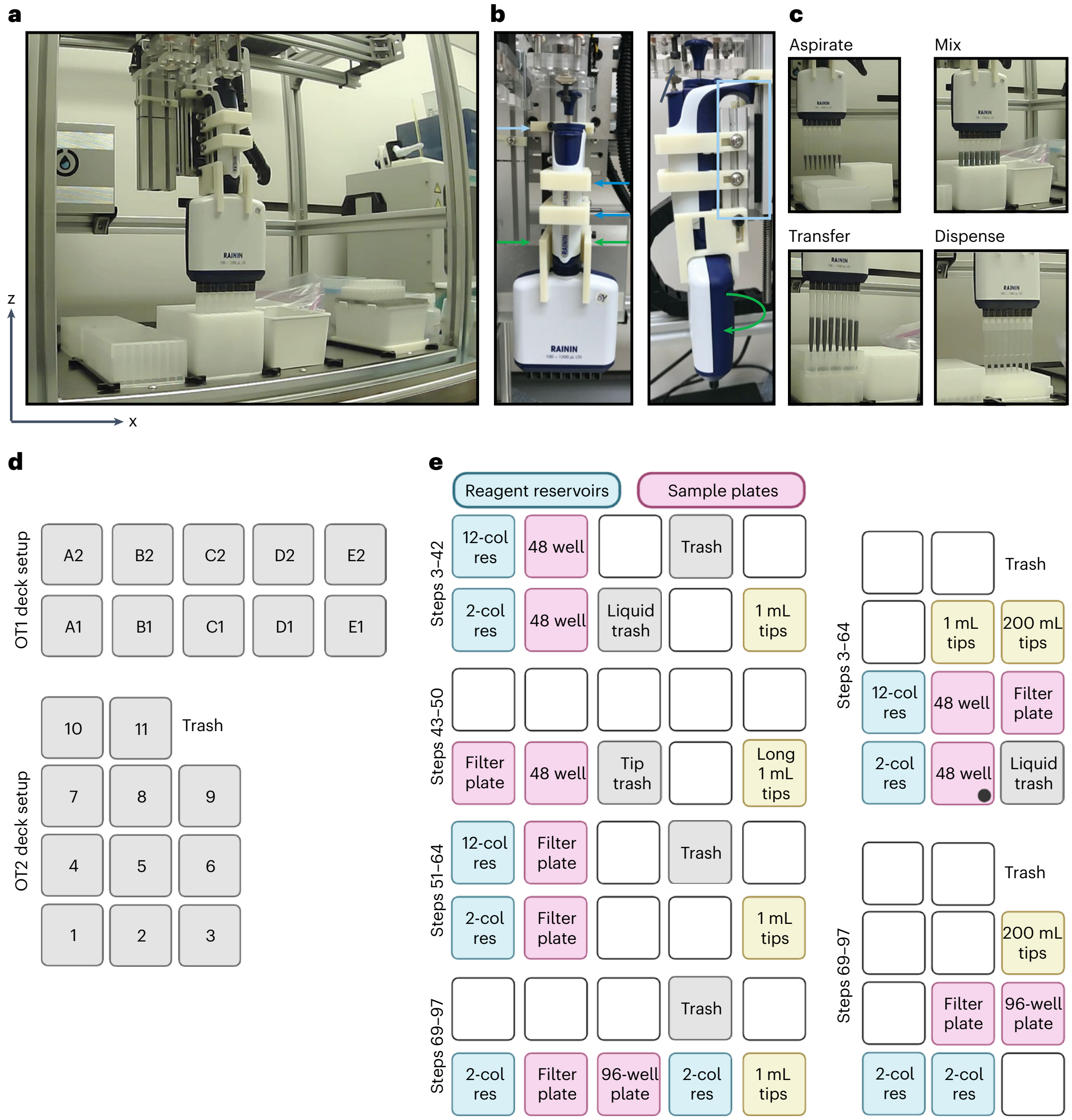

Fig. 2 | Overview of robotic system and comparison of setup using OT1 and OT2. a, Representative photograph of the full OT1 system. **b**, Close up of a mounted eight-channel high volume pipette on OT1. Dark blue, green and light blue indicators highlight 3D printed fixtures to prevent motion in the *x*, *y* and *z* directions, respectively. Black arrow indicates moving fixture used to eject tips. **c**, Representative freeze-frame photographs of various tasks performed on the robotic system. **d**, Representative images of deck configurations for OT1 and OT2. **e**, Representative deck layouts as the user steps through each section. Each deck layout describes the deck positions associated with a given consumable as consumables are calibrated per deck position. Note that a given consumable may move between positions as described in the procedure. The black circle highlights a position and consumable for which the user will have to calibrate the OT2 one-channel 1 mL pipette, in addition to the eight-channel 300 µL pipette. Col, column; res, reservoir; for example, 12-col res indicates a 12-column reagent reservoir. Figure created with BioRender.com.

https://support.opentrons.com/s/article/How-positional-calibration-works-on-the-OT-2. Unlike the OT1, two pipettes are required: the 300 µL eight-channel pipette and the single channel 1 mL pipette. Figure 2e indicates with a black circle the deck positions where the 1 mL pipette will be used in addition to the eight-channel 300 µL pipette.

Protocol overview and validation

The reagent volumes required to isolate cfRNA depend on initial sample volume (typically 1 mL for our process). Extracting cfRNA from 1 mL of plasma requires 5 mL of added reagents, yielding a maximum total volume of 6 mL. Since this is far beyond what a typical 96-well deep well plate can accommodate (~1–2 mL), we opted to use 48 deep-well plates to maintain system scalability (Fig. 1). Using a 48 deep-well plate not only interfaces well with the robotic system, but it also allows us to move away from the classic 50 mL conical tubes used for this step, since these can be difficult to process quickly in an automated fashion (Fig. 1a). Processing 96 samples using 48-well plates requires that the samples are split into two batches briefly and then recombined into one plate for subsequent steps (Fig. 1b). Specifically, we thaw and extract cfRNA from 48 samples (Fig. 1b, left side) and then refrigerate the isolated cfRNA while thawing and processing cfRNA from another 48 samples (Fig. 1b, right side).

We investigated whether this refrigeration step had adverse effects on cfRNA concentration by spiking in a known concentration of a single RNA oligonucleotide at the start of extraction. We then measured its concentration using RT–qPCR pre- and postrefrigeration as compared with that for control samples processed without any refrigeration (Fig. 3a, left). Across two independent trials, we find that refrigeration of isolated cfRNA for 1.5 h has no adverse effects on final yield (Bonferroni-adjusted P values 0.43 and 1.0 per trial, Wilcoxon signed-rank test) (Fig. 3a, right). Further, we find no loss in yield for samples postrefrigeration as compared with control unrefrigerated samples (Bonferroni-adjusted p values 1.0 and 1.0 per trial, one-sided Mann–Whitney rank test). We also separately confirmed that the custom RNA oligonucleotide and its corresponding TaqMan probes work as expected (Fig. 4).

Subsequently once cfRNA from all 96 samples has been extracted, all samples are treated with DNAse to remove any lingering DNA, then cleaned and concentrated to a final volume of 12 µL (Fig. 1b). In total, this process takes 4.5 h, with each sample at room temperature (15–25 °C) for 3 h at a given time (Fig. 1b). Finally, we quantified the likelihood of cross-contamination between samples by spiking in RNA only into every other well so that each 48-well plate then contained 24 sample wells and 24 blank wells.

Across two independent trials with 48 blank wells each (24 per 48-well plate), we find no evidence of cross-contamination (Fig. 3b). Specifically, we find that blank samples do not yield a distribution of RT–qPCR cycle thresholds (C_t) significantly lower than that of true nontemplate controls (NTC, $n = 17$) (Bonferroni-adjusted P values 0.16 and 0.10 per trial, one-sided Mann–Whitney rank test). Positive controls (PCs) are included for reference (Fig. 3b).

When setting up the system, and periodically when processing samples regularly, we recommend that the user confirms the absence of cross-contamination before processing precious clinical samples.

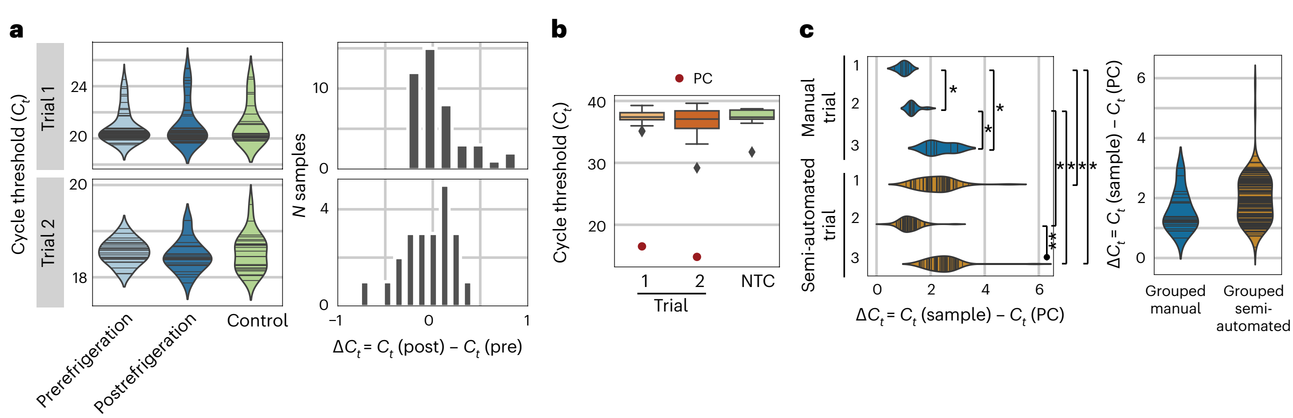

Fig. 3 | Semi-automated RNA isolation system validation using a synthetic RNA control. a, Comparison of RT–qPCR cycle thresholds (C_t) for paired samples pre- and postrefrigeration (light and dark blue) as compared with unrefrigerated samples (green) for two independent trials (rows). Histograms (right column) further highlight the distribution of difference in C_t between paired samples post- and prerefrigeration. **b**, Control experiment to identify any cross-contamination during processing. Yellow and orange box plots show the distribution of C_t for processed water samples from two independent trials as compared with C_t values for known NTC in green. Red points indicate PCs. Center line, box limits, whiskers and diamond shaped outliers represent the median, upper and lower quartiles, 1.5× interquartile range and any outliers outside that distribution, respectively. **c**, Comparison of manual and semiautomated cfRNA extraction with three trials per method. Top and bottom violin plots compare all trials in either an independent fashion or grouped together respectively. Symbols * and ** indicate statistically significant differences between the two indicated groups at P value thresholds less than of 0.05 and 10^{-10} respectively. P values were calculated using one-sided Mann–Whitney rank test with Bonferroni correction. For exact values, see the main text.

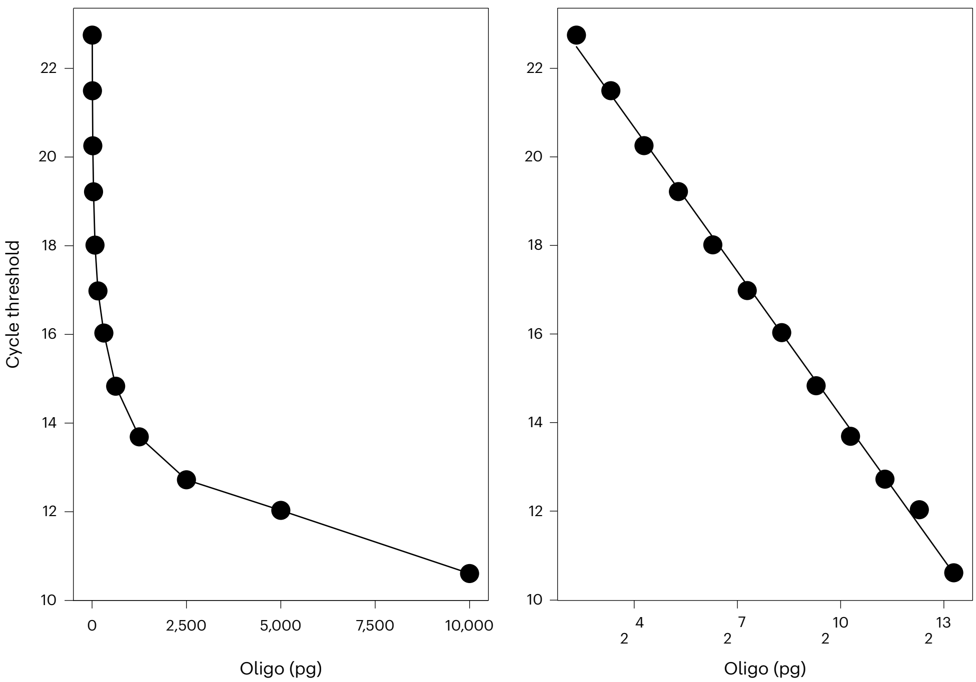

Fig. 4 | Validation of custom RNA control, ERCC54 and its corresponding TaqMan probes across a range of oligonucleotide input amounts. To confirm that the custom 120 RNA oligonucleotide is detected as expected by the corresponding Taqman probes (see 'Reagents'), we performed RT–qPCR across a 12-point two-fold dilution series. Left and right plots show linear and log x-axis values, respectively.

RT–qPCR validation

To compare the semi-automated pipeline with the gold standard of traditional manual pipetting, we assessed the concentration of RNA rescued for a known control relative to the concentration of RNA expected with total rescue (ΔC_t). Over three independent trials per protocol, we find that the manual and semi-automated methods perform comparably (Fig. 3c).

Globally, when we combined ΔC_t values from all three trials per protocol, we found that extracting RNA using the semi-automated method ($n = 144$, 48 per trial) did not significantly decrease RNA rescue when compared with the manual protocol ($n = 24$, 8 per trial) (Bonferroni-adjusted P value 0.11, one-sided Mann–Whitney rank test). However, when comparing individual trials in a pairwise fashion, we observed some significant shifts to more positive ΔC_t values, specifically between all three manual (M) trial pairs, one semi-automated (SA) trial pair and four of nine manual and semi-automated trial pairs (Bonferroni-adjusted P values 0.031, 0.0075, 0.016, 1.83×10^{-14}, 0.0001, 5.86×10^{-5}, 0.002, 0.00013 for pairs M1:M2, M1:M3, M2:M3, SA2:SA3, M1:SA1, M1:SA3, M2:SA1, M2:SA3, respectively, one-sided Mann–Whitney rank test). We note that because the manual trials ($n = 8$) require smaller batch sizes than their semi-automated counterparts ($n = 48$), pairwise comparison differences may be the result of subsampling rather than differences in protocol yield. The lack of a significant difference under global testing conditions supports this hypothesis.

Comparison with other methods

Cost, time and sustainability comparison with manual protocol

When compared with a manual protocol, assuming a batch size of eight samples (Fig. 1a), semi-automated parallel processing results in a 12-fold increase in sample batch size and nearly an 11-, 3- and 4-fold reduction in time, cost and biohazardous waste produced, respectively (Table 1). We note that this RNA isolation protocol, like many of its kind, lends itself well to parallelization because incubation and centrifugation compose nearly half of total processing time. As the sample number increases, such steps remain near constant in time. Further, reduced plastic usage, specifically by shifting to plate-based (versus individual tube) consumables and reduced tip usage, majorly contributed to decreased cost and biohazardous waste produced per sample processed.

Comparison with other RNA extraction protocols

When compared with other available protocols for semi- or fully automated RNA extraction and library preparation, ours provides a complementary option for those looking to extract RNA from bodily fluids and larger sample volumes as compared with existing techniques (Table 2). For example, Lazaro-Perona and colleagues describe a low-cost, fully automated protocol for liquid handling for RNA extraction at the microliter scale using a magnetic separation system in a comparable time frame to ours[21]. In contrast, Mamanova and colleagues describe a relatively more expensive protocol to extract RNA from single cells using a low-volume (nL scale), high-throughput semi-automated workflow[22]. Consequently, we believe that these protocols provide a nice complement to one another and cover a broad range of liquid volumes (nL to mL) and price points.

This protocol is well suited for laboratories that are beginning to process more blood samples, which we view as medium throughput (10^2–10^3 samples per month), but not enough to justify expensive liquid handling purchases and full robotic automation (10^4 samples per month). Our protocol requires user input at several steps to avoid purchasing more expensive robotics. Large biobanks might instead turn to purchasing more expensive and also more autonomous equipment such as the Hamilton or Bravo. Opentrons also offers a library preparation system (https://opentrons.com/products/workstations/ngs/). Therefore, a user might even consider using the Opentrons system from RNA extraction to library preparation seamlessly.

Advantages and limitations of the protocol

The key benefits of a semi-automated system include an increase in sample throughput by 12-fold and a reduction in time spent, cost and biohazardous waste produced by nearly 11-, 3- and 4-fold, respectively, as compared with manual extraction. Such quantitative benefits are coupled with several qualitative improvements. Specifically, a semi-automated

Table 1 | Comparison of manual and semi-automated RNA isolation protocols

Method	Batch size (*n*)	Time per batch (h)	Cost per sample (USD)	Tips used per sample (*n*)
Manual	8	4.5	41.25	26
Semi-automated	96	4.5	15.24	7

Table 2 | Comparison of automated RNA isolation and library preparation protocols

	Sample preparation protocol	Robotic system	Initial investment (USD)	User involvement	Volume range	Batch size (*n*)	Time
This protocol	RNA extraction from blood plasma	OT1	5 K	Semi-automated	µL–mL	96	4.5 h
This protocol	RNA extraction from blood plasma	OT2	12 K	Semi-automated	µL–mL	96	4.5 h
Lazaro-Perona et al.[21]	RNA extraction from nasopharyngeal swabs	OT2	12 K	Fully automated	µL	96	3.5 h
Mamanova et al.[22]	RNA extraction from single cells	Zephyr G3, Mantis	100 K	Semi-automated	nL–µL	96	3 d
OT2 NGS workstation	Library preparation from cDNA	OT2	30 K	Semi-automated	µL	96	1–2 d

protocol allows for consistent sample preparation quality across users and reduced personnel time during the protocol. When contrasted with fully automated systems, we find that a semi-automated system is well suited to the needs of a typical academic laboratory performing discovery research. Although a semi-automated system requires the user to remain nearby, its initial cost (US$5.5 K) remains substantially cheaper than that of its automated counterparts (US$20–100 K).

Importantly, the system's low cost and easy user-level customization can be applied to automate any RNA isolation required for discovery research in an academic laboratory. As the system is specifically built to leverage existing 96-well kits for RNA isolation, cleaning and concentration, such a protocol can easily be transferred to isolate RNA from other sources such as tissues, urine or other bodily fluids, with relatively few changes. The user would only have to program the new liquid handling steps or modify existing ones using a common high-level programming language (Python). This easy customization stands in stark contrast to the proprietary software or lower-level programming languages such as C, which is less user friendly, required to modify protocols on most automated systems.

The semi-automated protocol we describe is not without its limitations. The OT1 system does not permit for easy pipette changes, restricting the user to the volume range of a single multichannel pipette. In an academic laboratory setting, we do not see volume restrictions as a major issue since most RNA protocols require reagent volumes within the range of a single pipette. The user can perform any steps that require reagent volumes outside the range of the mounted pipette by repeatedly transferring a smaller volume within the upper limit for larger volumes or transferring by hand for smaller volumes. We find performing the three steps that require smaller, variable volumes by hand provides additional flexibility in the protocol. For instance, adding samples by hand easily accommodates various sample storage strategies such as microtubes and matrix tubes. Additionally, adding the eluant by hand permits the user to easily modify the volume used for each RNA isolation to match their chosen downstream application.

We also found that setting up and calibrating manual pipettes to work with this platform required substantial effort. These pain points have since been addressed in the second generation of this robotic system, the OT2 (US$12 K including pipettes), which can be used in place of the OT1 system described here. The Python Application Programming Interface to interface with the OT2 as compared with the OT1 may be slightly different and some function calls may need to be modified (https://support.opentrons.com/s/article/Modifying-existing-OT-One-protocols). The general liquid handling logic flow, however, will remain consistent, and we expect these changes to be minor. Although the OT1 has been phased out since its use in this work, users can choose to build their own system following directions described in 'Equipment setup'. Importantly, all necessary hardware and software required to independently build the system have been made available on GitHub (https://github.com/Opentrons/otone_hardware). Further, both OT1 and OT2 systems are open source, permitting users to easily and inexpensively customize either system to suit their needs.

Overall, we have shown that semi-automated sample processing can be performed affordably and quickly within individual academic laboratories. Shifting to semi-automated sample processing alleviates a major bottleneck in discovery-focused work and allows for seamless transitions from sample collection to data generation and analysis. With this robust and scalable solution, we can test hypotheses faster and spend less time on repetitive but important tasks such as sample processing.

Applications

The current protocol is suitable for plasma samples (ideally fresh, or previously frozen at −80 °C) or serum samples. An example bash script (https://github.com/miramou/cfRNA_pipeline_automation/blob/master/Opentrons/example/run_cfRNA_pipeline.sh) to launch the protocol is provided for 1 mL volume samples. Note that the volume of lysis buffer and ethanol (EtOH) added will depend on the initial sample volume and is defined using µL in the script using the variables, LYSIS_VOL and ETOH_VOL, respectively. If you would like to process a smaller initial sample volume, please edit the variables LYSIS_VOL and ETOH_VOL accordingly to reflect the appropriate reagent volume. Follow the instructions provided by Norgen, which manufactures the RNA extraction kit used here, to calculate the required volume of lysis buffer and EtOH.

General considerations

This protocol details how to extract cfRNA simultaneously from 96 samples using a robotic system. See https://youtu.be/g6RsSaNvSNA for a full run through. We provide resources to guide users to set up and calibrate a robotic system including a list of parts, specific scripts for calibration, and photographs and videos of a representative well-calibrated system. Manual pipetting can be substituted for robotic pipetting and may be preferred when learning or troubleshooting the protocol.

Our protocol is written for researchers with some familiarity with the command line and programming languages such as Python and Bash. Once the launch script starts and throughout the procedure, the system will prompt the user about when and where to add reagents, swap tip boxes and place consumables, among other necessary cues. Note that the stated reagent volumes at Steps 16 and 22 reflects the processing of 1 mL plasma samples.

Before processing samples, it is advised to map out the plate and randomize samples across it. If samples from a case/control study are processed, confirm that each half plate (48 samples) contains samples from both case and control subjects to ensure that any batch effect can be decoupled from the case effect.

This protocol describes both the OT1 and OT2 adapted versions. The most notable difference between the two is that the OT1 uses a single pipette, an eight-channel 1mL pipette, whereas the OT2 uses both an eight-channel 300 µL pipette (200 µL max volume with filter tips) and a single channel 1 mL pipette. This results in liquid handling differences for the OT1 as compared with the OT2 procedure, but the overall flow remains the same. Consequently, the user will have to modify existing OT1 scripts slightly to match the pipetting modifications described for OT2 throughout the procedure (https://support.opentrons.com/s/article/Modifying-existing-OT-One-protocols). Deck positions also differ between the OT1 and OT2 and are indicated as X, Y for OT1, OT2, respectively. Refer to Fig. 2 for a visual overview of the deck setup.

Considerations about the laboratory facilities

All the steps in our protocol should be done in a standard laboratory with a chemical fume hood. Before starting, care should be taken to ensure that that the workspace and all consumables placed within it are clean (e.g., using 70% EtOH) and RNase-free (e.g., using RNase Away).

Materials

Biological materials

- Fresh or frozen plasma samples
  ▲ **CRITICAL** Ensure that sample collection adheres to your local institutional review board or administrative panel on laboratory animal care guidelines. Tubes used for plasma isolation can contain most common solutions except for heparin, which inhibits reverse transcription downstream. More details about best practices can be found in[23] (see the section 'Experimental challenges when isolating circulating RNA').

Reagents

- Norgen Plasma/Serum Circulating and Exosomal RNA Purification 96-Well Kit (Slurry Format) (Norgen, cat. no. 29500)
- Baseline-ZERO DNAse (Lucigen, cat. no. DB0715K)
- Zymo RNA Clean and Concentrator-96 kit (Zymo Research, cat. no. R1080)
- RNA 6000 Pico Kit (Agilent, cat. no. 5067-1513)
- 2-Mercaptoethanol (Sigma-Aldrich, cat. no. M6250)

▲ **CAUTION** 2-Mercaptoethanol is highly corrosive and toxic and poses a health and environmental hazard. Avoid skin contact or inhalation. Wear full personal protective equipment when handling and always use within a chemical hood. Apply other safety measures if necessary and adhere to local guidelines when disposing.
- 200-Proof EtOH (Sigma-Aldrich, cat. no. E7023)
- Nuclease-free H_2O (IDT, cat. no. 11-04-02-01)
- iTaq Universal Probes One-Step Kit (Bio-Rad, cat. no. 1725141)
- SMARTer Stranded Total RNAseq Kit v2 – Pico Input Mammalian Components (Takara, cat. no. 634419)
- SMARTer RNA Unique Dual Index Kit – 96U Set A (Takara, cat. no. 634452)
- 120 nt custom RNA oligonucleotide that matches the 3′ end of External RNA Controls Consortium 54 (ERCC54) (IDT, custom RNA oligonucleotide):
  UUU AGA AUG CUU AAA GAU GGC AGA GUU GGA GGA GAG AUU UGC CAA UCA CAA
  ACC AAA UCA GUU GAG UGG AGG GCA ACA ACA GAG AGU UGC UAU AGC GAG GGC
  UUU GGC AAA CAA CCC ACC
- TaqMan gene expression assay for ERCC54 (Thermo Fisher, cat. no. 4448490, Assay ID Ac03459999 a1)

Automation compatible consumables
- 48-Well deep well plates (Thomas Scientific, cat. no. 1149Q15)
- 12-Column reagent reservoir (Thomas Scientific, cat. no. 1149Q70)
- Two-column reagent reservoir (Thomas Scientific, cat. no. 1149Q79)
- 96-Well PCR plate (Bio-Rad, HSP9631)
- 2 mL deep-well plates (Thomas Scientific, cat. no. 1149J85)
- Razor blades (Ted Pella, cat. no. 121-32)
- Ziploc gallon bags (SC Johnson, cat. no. 665256)
- Aluminum foil seals (Sigma-Adrich, cat. no. Z722634)
- Adhesive clear plate seals (Fisher Scientific, cat. no. AB-0558)

Equipment
Common equipment
- 2100 Bioanalyzer instrument (Agilent, cat. no. G2939BA)
- Eight-channel, 20 μL pipette, L8-P20 (Rainin, cat. no. 17013803)
- Vortex (Fisher Scientific, cat. no. 02-216-100 and cat. no. 14-955-163)
- Incubator (Fisher Scientific, cat. no. 07-201-060)
  ▲ **CRITICAL** Incubator must reach 60 °C.
- Centrifuge (Thermo Scientific, cat. no. 75009911 and 75003017)
  ▲ **CRITICAL** Centrifuge must achieve spin speeds of 3,000–5,000*g* and include buckets for deep well plates.
- −80 °C and −20 °C freezer
- Refrigerator
- Real-Time PCR Detection System (Bio-Rad, CFX96 or CFX384, cat. no. 1845096 or 1855484)

OT1 procedure-specific equipment
- Opentrons 1.0 robot or home-built OT1 system (see ‘Equipment setup’)
- 3D printer
- 3D printed parts (https://github.com/miramou/cfRNA_pipeline_automation/tree/master/Opentrons/custom_solidworks_parts)
- Eight-channel, 1.2 mL pipette, L8-P1200 (Rainin)
- 1.2 mL pipette tips (Rainin, cat. no. 30389231)
- 1 mL extended length pipette tips (Rainin, cat. no. 30389223)
  ▲ **CRITICAL** Extended length tips are necessary to ensure that the full sample volume is transferred at Step 48. If substituting for another pipette and its corresponding tips, one should ensure that the tips used are adequately long to reach the bottom of a 48-well deep well plate.

OT2 procedure-specific equipment

- Opentrons 2.0 robot
- Eight-channel, 300 µL pipette (Opentrons, GEN2). Note that this pipette can hold a maximum volume of 200 µL with filter tips.
- Single channel, 1,000 µL pipette (Opentrons, GEN2)
- 200 µL filter tips (Opentrons)
- 1,000 µL filter tips (Opentrons)

Software

- Opentrons app (https://opentrons.com/ot-app/)
  - OT1: v2.5.2
  - OT2: v6.3.0
- Miniconda (https://docs.conda.io/en/latest/miniconda.html)
- Python
- Bash or equivalent shell and command language
- Software for running the protocol is available at https://github.com/miramou/cfRNA_pipeline_automation/tree/master/Opentrons
- Software for processing RNA-sequencing data is available at https://github.com/miramou/cfRNA_pipeline
- Software to reproduce the figures in this manuscript is available at https://github.com/miramou/cfRNA_pipeline_automation/tree/master/Manuscript_Figs

Reagent setup

Exogenous RNA control

Reconstitute lyophilized 120 nt custom RNA oligonucleotide in nuclease-free water as per the manufacturer's instructions, at a concentration of 1 µg/µL. On ice, aliquot stock solution into 2.5 µL single-use aliquots. Store at −80 °C indefinitely.

Lysis buffer A (Norgen)

Add 1.2 mL 2-mercaptoethanol per 100 mL lysis buffer. Store at room temperature for up to 1 year.

Wash buffers (Norgen and Zymo)

Add 200-proof EtOH to wash buffer per manufacturer instructions. It is best to make this the day of or day before; however, it can be stored at room temperature for some time. We have tested the protocol with wash buffer made earlier in the month successfully.

Lysis buffer and slurry

Preheat in a 60 °C incubator, vortexing occasionally until all precipitate dissolves and slurry appears well combined.

▲ **CRITICAL** Residual precipitate may clog pipette tips during reagent transfer, leading to inaccurate pipetting. Keep the buffer and slurry heated until immediately before use, at which point, vortex once more.

Equipment setup

General setup

- Filter plate balance (Norgen, Zymo) and 48-well-plate balance: ensure that you have balances for each of these plate types, which are used in several centrifugation steps
- Clean and preheat incubator to 60 °C
- Clean all work surfaces including OT1 or OT2 robot deck, the surrounding workspace in the chemical hood and labware with 70% EtOH and a RNase decontaminant to ensure a RNase-free space.

Building the OT1

- Part designs to completely reconstruct the OT1 are available at this link: https://github.com/Opentrons/otone_hardware

- Follow the protocol at this link (http://pipettejockey.com/2018/01/03/making-a-opentrons-compatible-liquid-handling-robot/ or see Supplementary Manual) that describes how to build the OT1 from scratch
- For troubleshooting, refer to this video (https://youtu.be/BhQub5Xh_8o) that describes common pitfalls when building the OT1

Software setup

- Follow the setup instructions provided under the heading 'Getting started' at this link[24]: https://github.com/miramou/cfRNA_pipeline_automation/tree/master/Opentrons
- Clone the repository and create a new miniconda environment using the provided requirements file, as detailed in the link provided: https://github.com/miramou/cfRNA_pipeline_automation/blob/master/Opentrons/env/ot.yaml

Procedure

Prepare samples

● **TIMING 10 min**

1. Before thawing any samples, ensure that you have mapped out the plate (see 'General considerations'). Of special note, if samples are associated with a case/control, confirm that each half plate (48 samples) contains samples from both case and control subjects to ensure that any batch effect can be decoupled from the case effect.
2. See option A or B for compatible sample types that we have tested. We strongly recommend starting with option A as the user familiarizes themselves with the procedure. Once the user feels comfortable with the entire approach, option B becomes the standard. Given the precious nature of clinical samples, we have found it important to thoroughly test the entire procedure first using the RNA control you prepared previously (see 'Reagent setup').
   (A) **RNA control**
      (i) On ice, thaw the single-use aliquot of short 120 nt RNA control that you prepared previously.
      (ii) Once thawed, quickly spin down and dilute further to yield a solution at 5 ng/µL by combining 1 µL of stock with 199 µL of nuclease-free water.
      (iii) To mimic the expected sample volume of 1 mL, pipette 1 mL of nuclease-free water per well across 48 wells of a 96-deep-well plate on ice.
      (iv) Add 1 µL of diluted stock to each well of the stock plate and mix well by pipetting. Seal and store at 4 °C until needed for RNA extraction.
   (B) **Frozen plasma samples**
      (i) At room temperature, thaw the first 48 plasma samples. Do not thaw at 4 °C to prevent the formation of cryoprecipitates that inhibit RNA extraction performance.
      (ii) Organize samples to match the planned plate layout so that they can be easily pipetted using a multichannel pipette (RNA extraction Step 17).
      (iii) As samples thaw, proceed to preparatory steps for RNA extraction (Steps 3–16).
      (iv) It is good practice to include at least one positive and one negative control per 48-well plate (46 samples and 2 controls). Use the instructions from Step 2A to prepare your PC. The negative control can be 1 mL of nuclease-free water.

RNA extraction

● **TIMING 3 h (1.5 h per 48-well plate)**

3. Turn on and connect the robotic system. Confirm that the computer recognizes the robotic system using the Opentrons app. Home the system by pressing the home button on the app. Homing the system is a step that enables the robotic arm to return to its initial, home position and ensures that the robotic arm and code are in sync.

4. Using the shell, deploy the launch script titled run 'cfRNA pipeline.sh' passing one argument that specifies the folder where you would like log files saved as indicated by <expt_prefix> below.

```
bash cfRNA_pipeline.sh <expt_prefix>
```

▲ **CAUTION** The sample launch script is programmed to process 1 mL samples. Other sample volumes require different reagent volumes. Refer to the manufacturer's instructions (Norgen Plasma/Serum Circulating and Exosomal RNA Purification 96-Well Kit) for these volumes and modify global variables LYSIS_VOL and ETOH_VOL in the launch script accordingly.
5. Seal one new two-column reagent reservoir and one new 12-column reagent reservoir using a clear plate seal. The seal will be sequentially cut to prevent reagent cross-contamination.
6. Using a razor blade, cut the seal on the 12-column reservoir such that the boat labeled 1, which will be closest to the user once placed on the deck, is now exposed and part of boat 2 immediately adjacent to it is also open. Reagents will only be added to every other boat, specifically boats with odd numbers, to further prevent reagent cross-contamination and prevent pipette tips from sticking to the seal when entering a boat.
7. Using a razor blade, cut the seal on the two-column reservoir such that the boat labeled 1, which will be closest to the user once placed on the deck, is now exposed.
8. Place the tip boxes on the deck.
   - For the OT1, place a standard length 1 mL filter tip box (gray if using Rainin) at position E1 and completely remove the lid.
   - If using a pipette box like Rainin's whose footprint does not match that of deck slot, 3D print and use a pipette box holder to fix the pipette box's position. This is not necessary for the OT2 system that uses automation-compatible tip boxes. For Rainin, we provide a box holder design at https://github.com/miramou/cfRNA_pipeline_automation/blob/master/Opentrons/custom_solidworks_parts/tip_box/OPENTRONS_RAININ_TIP_BOX_HOLDER_2.STL
   - For the OT2, place the 200 µL filter tip box at position 9 and 1 mL filter tip box at position 8.
9. If you are using the OT1 system, place a Ziploc bag with the sides rolled down halfway at position D2. This will act as trash where the robot will dispose of used tips. Ensure that the bag sits upright. The OT2 system has a fixed trash position that will be used and does not require a Ziploc bag.
10. Place the two-column reagent reservoir at position A1,1. Place the 12-column reagent reservoir at A2,4.
11. Place a new 48-deep-well plate at position B2,5.
12. Vortex preheated slurry solution from the Norgen kit well and add 10.5 mL to column 1 (closest to the user and fully unsealed) in the 12-column reservoir at A2,4.
    ▲ **CRITICAL STEP** Proceed quickly. In our experience, cooled slurry crystallizes and clogs the pipette tip upon reagent transfer.
13. Press enter to proceed with procedure. The robotic system will now pick up pipette tips and add 200 µL slurry to each row of the 48-well plate and then dispose of used tips.
    - For the OT1, the robot will pick up 1 mL tips using the eight-channel 1 mL pipette.
    - For the OT2, the robot will pick up 200 µL tips from position 9 using the eight-channel 300 µL pipette.
      ◆ **TROUBLESHOOTING**
14. Move the 48-well plate to position B1,2 once the system prompts you.
15. Vortex the preheated lysis buffer A from Norgen kit and add 110 mL to position 1 (closest to the user and unsealed) in the two-column reservoir at A1,1. To avoid buffer crystallization in the OT2-compatible version, which requires nine liquid transfers (Step 16), we recommend adding ~40 mL lysis buffer to the reservoir at any given time.
    ▲ **CRITICAL STEP** Proceed quickly. In our experience, cooled lysis buffer crystallizes and clogs the pipette tip upon reagent transfer.

16. Press enter to proceed. The robotic system will now pick up pipette tips and add 1.8 mL lysis buffer to each row of the 48-well plate and then dispose of tips.
    - For the OT1, this will be done in two serial dispenses per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in nine serial dispenses per row using the eight-channel 300 µL pipette.
17. Using a new 1 mL filter tip box and multichannel 1 mL pipette, transfer samples by hand (see option A or B in Step 2) to the 48-well plate and mix each column by pipetting up and down at least five times. Set aside the tip box. You will use the same box for all 96 samples in two rounds of RNA extraction.

    ▲ **CRITICAL STEP** Do not use an already opened tip box on the robot. The protocol is programmed to know how many rows of tips have been used and pick up where it left off. Additionally, if processing 96 samples, a single tip box can be entirely dedicated to sample transfer and set aside between batches of 48 samples.
18. Seal plate well and incubate at 60 °C for 10 min in the preheated incubator (see ‘General setup’ under ‘Equipment setup’).
19. Type ‘y’ and press enter to proceed.
20. Once there is ~1 min left of the incubation, remove the seal completely from the two-column reservoir at position A1,1 and add 150 mL 200-proof EtOH to the second column further from the front.
21. After the incubation is complete, remove the seal from the 48-well plate and place it at deck position B1,2.
22. Press enter to proceed. The robotic system will now pick up pipette tips and add 3 mL 200-proof EtOH to each row of the 48-well plate and then dispose of used tips.
    - For the OT1, this will be done in three serial dispenses per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in 15 serial dispenses per row using the eight-channel 300 µL pipette.
23. The system will now automatically proceed to mixing each row of samples by pipetting up and down at different heights and dispose of tips after each row to ensure that each 5 mL sample is well mixed.
    - For the OT1, this will be done per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done per well using the single-channel 1 mL pipette.
24. Once prompted, seal the 48-well plate well and centrifuge for 1 min at 1,000*g* at room temperature to pellet the slurry. Ensure that the centrifuge is appropriately balanced. Move the 48-well plate back to the chemical hood.
25. Place a hard-sided open vessel at position C1,3. This will serve as liquid trash as the robot decants supernatant from the 48-well plate. Typically, an empty tip box with the tip rack removed works well for this purpose.

    ▲ **CRITICAL STEP** When performing the procedure for the first time, ensure that pipette tips do not touch the liquid waste when decanting and that splatter is minimized. Recalibrate the labware’s position on the OT deck otherwise (see ‘Robotic system calibration’ in the ‘Experimental design’ section for more detail).
26. We will remove the seal sequentially from each row. Using a razor blade, cut the seal between columns 1 and 2 and remove the seal to reveal the eight samples in column 1 alone. Place the plate at position B1,2 with column 1 closest to the user.
27. Press enter to proceed. The robot will now decant all supernatant from column 1 and then dispose of tips. If at any point the trash bag containing tips becomes full, use the programmed pause between columns to replace the trash bag with a new one.
    - For the OT1, decanting will be done per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done per well using the single-channel 1 mL pipette.

      ▲ **CAUTION** Dispose of biohazardous trash and 2-mercaptoethanol according to local environmental guidelines.

      ◆ **TROUBLESHOOTING**

28. Repeat Steps 26 and 27 for each column of the 48-well plate.
    - For the OT1, after the third column, the system will prompt you to replace the now empty tip box at E1 with a new tip box. Do so, taking care to remove the plastic lid completely from the tip box. As before, press enter to proceed.
    - For the OT2, replace the 1 mL tip box located at deck position 8 after all wells have been decanted.
29. Having decanted all supernatant, move the 48-well plate to position B2,5.
30. Remove the liquid waste container from the deck and set aside for now.
31. For the 12-column reservoir at position A2,4, remove the seal from column 3 and part of column 4 and add 20 mL heated lysis buffer to column 3 as prompted by the system. Place the reservoir back at A2,4.
32. Press enter to proceed. The robot will now add 300 µL of lysis buffer to each row of the 48-well plate and dispose of used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in two transfers per row using the eight-channel 300 µL pipette.
33. Seal the 48-well plate. Vortex for at least 30 s, passing the plate bottom across the vortex to ensure that all samples are well mixed.
34. Incubate the 48-well plate for 10 min at 60 °C.
35. During the incubation, prepare the filter plate included in the Norgen kit and place it on top of a 2 mL 96-well collection plate. If this is the first 48-well sample batch, seal the entire filter plate and cut between rows 6 and 7 with a razor. Rows 7–12 will remain sealed until processing the next batch of 48 samples. Set this aside for now.
36. Once there is ~1 min left from the incubation, remove the seal from column 5 and part of column 6 from the 12-column reservoir at position A2,4 and add 20 mL 200-proof EtOH to column 5.
37. Remove the seal and place the 48-well plate back at position B2,5.
38. Press enter to proceed. The robot will now add 300 µL of 200-proof EtOH to each row of the 48-well plate and dispose of used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in two transfers per row using the eight-channel 300 µL pipette.
39. As the robot proceeds, change the incubator set point as follows:
    - Processing a total of 96 samples, following the first set of 48 samples: keep the incubator at 60 °C.
    - Processing a total of 96 samples, following the second set of 48 samples: set the incubator to 37 °C for DNase incubation in next section.
    - Processing a total of 48 samples, following the first set of 48 samples: set the incubator temperature to 37 °C for DNase incubation in next section.
40. Seal the 48-well plate and vortex for 1 min, passing the plate bottom across the vortex.
41. For the OT1, remove the regular length tip box at position E1, place the lid back and set the tip box aside. You will use the tip box again in Step 51.
42. Prepare the system for sample transfer from the 48-well plate to the filter plate.
    (A) **For the OT1**
        (i) Place an extended length tip box (blue for Rainin) without the lid at position E1. It is important to use extended length tips (102 mm long) as regular length tips do not reach the bottom of the 48-well plate.
    (B) **For the OT2**
        (i) Place a full standard-length tip box at position 8 without the lid.
        (ii) For the first set of 48 samples: use a new tip box.
        (iii) For the second set of 48 samples: use the same tip box as in 42B(ii). Make sure that pipette box is positioned such that the six remaining tip rows are closest to you when facing the system.
43. Remove the two-column reservoir at A1,1 and dispose of it.

44. Prepare the trash position.
    - If using the OT1, place an empty tip box at position C1. This will act as trash for this step and the system will dispose of used pipette tips here. Ensure that tips fall flat and do not obstruct the motion of the robot when they are discarded.
    - If using the OT2, the system will use the fixed trash position to discard used tips. Empty this fixed position.
45. Place the Norgen filter plate you prepared in Step 35 at position A1,6 with the corner well A1 closest to you on the left when facing the system.
    ▲ **CRITICAL STEP** Often there is a little wiggle room in how the filter plate sits on top of the collection plate. Take care to always position the filter plate in the same spot relative to the collection plate before running the procedure so that the pipette's saved calibration remains true. We recommend pulling the filter plate as far up and to the left when facing the system as the collection plate allows each time.
46. The system will now prompt you and ask what filter row to start at.
    - For the first set of 48 samples: type 1 and press enter.
    - For the second set of 48 samples: type 7 and press enter.
47. Vortex the 48-well plate again focusing on the first column. Remove the seal from that column alone using a razor blade as before. Place the plate at position B1,5.
    ▲ **CRITICAL STEP** Move quickly after vortexing so that the slurry mixture does not settle again.
48. Press enter to proceed. The system will now pick up long pipette tips, aspirate the sample and reagents present in column 1 of the 48-well plate in B1,5, transfer them to column 1 of the 96-well filter plate in A1,6 and dispose of used tips in C1,Trash.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in one transfer per well across the row using the single-channel 1 mL pipette.
49. Repeat Steps 47 and 48 for each row of the 48-well plate.
50. Centrifuge the filter and collection plates with sample rows unsealed (i.e., rows 1–6 for first 48-well plate and then rows 7–12 for second) for 4 min at 2,500–4,000*g* at room temperature.
    ◆ **TROUBLESHOOTING**
51. Prepare the deck for washes.
    - For the OT1, place the regular length tip box (gray for Rainin) set aside at Step 41 at position E1 in the same orientation (i.e., empty rows facing the front). The system knows to start picking up tips at row 6. Set aside an extended length filter tip box with a lid (blue for Rainin); you will use these extended length tip rows for the next set of 48 samples.
    - For the OT2, set aside the box of 1 mL pipette tips at position 8; you will use these for the next set of 48 samples. You will proceed for the next steps with the eight-channel 300µL.
52. Following the system's prompting, place a two-column reagent reservoir at position A1,1. Place the reservoir in A1,1 such that the open, clean reservoir boat is closest to the front of the system and user.
    - For the first set of 48 samples: use a new two-column reservoir. Like before, seal both reservoir boats and then remove the seal only from one.
    - For the second set of 48 samples: use the same reservoir as for the first set of samples. Make sure to flip it such that the previously sealed, clean boat is now facing the front.
53. Place the filter plate with the collection plate underneath at position B1,6.
    - For the first set of 48 samples: filter plate column 1 should be closest to the front of the system and user.
    - For the second set of 48 samples: filter plate column 12 should be closest to the front of the system and user.
54. Add 24 mL of Norgen wash buffer to the open reservoir boat in the two-column reservoir at A1,1.

55. Press enter to proceed. The system will now add 400 µL of wash buffer to the six rows closest to the front and dispose of the used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in two transfers per row using the eight-channel 300 µL pipette.
56. Centrifuge the unsealed filter plate with collection plate underneath for 3 min at 2,500–4,000*g* at room temperature, taking care to ensure that the system is balanced. Discard flow through.
57. Repeat Steps 53–56 twice more for a total of three washes.
58. Centrifuge the filter and collection plate with sample rows unsealed (i.e., rows 1–6 for first 48-well plate and then rows 7–12 for second) for an additional 5 min at 2,500–4,000*g* at room temperature to ensure that the filter plate is fully dry.
59. Replace the collection plate underneath the filter plate with the elution plate. Take care to match positions between the filter and elution plate such that filter position A1 sits on top of elution plate position A1.
    - For the first set of 48 samples: use a new, labeled 96-deep-well plate.
    - For the second set of 48 samples: take the previously used elution plate out of the fridge, remove the seal and place it underneath the filter plate.
60. For the 12-column reservoir at position A2,4, remove the seal from column 9 and part of column 10 and add 10 mL elution buffer to column 9 as prompted by the system. Place the reservoir back at A2,4.
61. Place the filter plate with the elution plate underneath at position B2,6.
    - For the first set of 48 samples: filter plate column 1 should be closest to the front of the system and user.
    - For the second set of 48 samples: filter plate column 12 should be closest to the front of the system and user.
62. Press enter to proceed. The system will now add 120 µL of elution buffer to the first six rows and dispose of the used tips. In our hands, adding 120 µL of elution buffer results in a final elution volume of 100 µL.
63. Centrifuge the filter and elution plate with sample rows unsealed (i.e., rows 1–6 for first set of 48 samples and then rows 7–12 for second set of 48 samples) for 5 min at 2,500–4,000*g* at room temperature.
64. Confirm that six sample columns in elution plate now contain 100 µL of flow through and eluted cfRNA. Select Option A, B or C depending on the number of samples processed and the stage they are at:

    (A) **Processing a total of 96 samples, following the first set of 48 samples**
      (i) Move the filter plate to a new collection plate and set aside keeping columns 7–12 sealed.
      (ii) Seal the elution plate that now contains eluted samples for columns 1–6 and place in the fridge.
      (iii) Set aside a tip box at E1,9 (1 mL tips for OT1, 200 µL tips for OT2) with three rows remaining. You will use these in the 'RNA cleaning and concentration' section.
      (iv) Enter 0 at the system's prompting when asked whether to 'Proceed to Zymo?', which will end the script. Repeat Steps 4–63 for the second set of 48 samples.

    (B) **Processing a total of 96 samples, following second set of 48 samples**
      (i) Enter 1 at the system's prompting when asked whether to 'Proceed to Zymo?'.
      (ii) Set aside a tip box at E1,9 (1 mL tips for OT1, 200 µL tips for OT2) with three rows remaining. You will use these in the 'RNA cleaning and concentration' section.
      (iii) Proceed to the next section, 'DNase digestion'.

    (C) **Processing a total of 48 samples**
      (i) Enter 1 at the system's prompting when asked whether to 'Proceed to Zymo?'.
      (ii) Proceed to the next section, 'DNase digestion'.

DNA digestion

● **TIMING** 20 min

65. Prepare a master mix of DNase and 10× DNase buffer. Each sample requires 11 µL of buffer and 2 µL of DNase. For 96 samples, combine 1,320 µL buffer and 240 µL DNase enzyme. Mix well and keep on ice until ready.
66. Dispense 13 µL of DNase master mix by hand to each sample well in the elution plate (Step 64) using a P20 multichannel pipette and a disposable reagent reservoir.
67. Seal the plate and mix by vortexing lightly.
68. Incubate for 20 min at 37 °C in the incubator you preheated in Step 39. During the incubation, proceed to Steps 69–72.

RNA cleaning and concentration

● **TIMING** 1 h

69. Place a tip box at position E1,9. If you are processing a total of 96 samples, use the tip box you set aside in Step 64 with three rows remaining facing the front. If you are processing a total of 48 samples, use a new tip box.
70. Seal and place two new two-column reagent reservoirs at positions A1,1 and D1,2.
71. Remove the seal from the column closest to the front of the chemical hood for the D1,2 reservoir.
72. Prepare the filter plate included in the Zymo kit and place it on top of a 2 mL 96-well collection plate. Set this aside for now.
73. Once the incubation is complete, place the sample plate from Step 68 at position C1,6.
74. Enter either 6 or 12 at the system's prompting to indicate the final row number to be processed, where 6 would indicate 48 samples and 12 would indicate 96 samples.
75. As prompted by the system, add 27 mL binding buffer to the unsealed reservoir column in position D1,2.
76. Press enter to proceed. The system will now add 226 µL of binding buffer to the number of specified rows and dispose of the used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in two transfers per row using the eight-channel 300 µL pipette.
77. As prompted by the system, remove the seal for the second reservoir column in position D1,2 and add 50 mL 100% EtOH.
78. Press enter to proceed. The system will now add 339 µL of EtOH to the number of specified rows and dispose of used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in two transfers per row using the eight-channel 300 µL pipette.
79. If processing 96 samples, change the tip box for a new tip box as prompted by the system.
80. Place the filter plate prepared at Step 72 at position B1,5.
81. Ensure that corner A1 for both the sample plate at C1,6 and filter plate at B1,5 is top left closest to the user when facing the system.
82. Press enter to continue. The system will now sequentially mix each row of samples and then transfer the full sample volume to the corresponding filter plate row switching tips between rows.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in four transfers per row using the eight-channel 300 µL pipette.
83. Once all samples have been successfully transferred to the filter plate, centrifuge the now-filled filter plate for 5 min at a speed between 3,000 and 5,000*g* at room temperature. Ensure that the centrifuge is balanced. Dispose of the flow through and reassemble the filter and collection plate in the same orientation.
84. If processing 96 samples, replace the now empty tip box at E1,9 with the second tip box you set aside in Step 64 that has three rows remaining facing the front.

85. Add 48 mL of RNA preparation buffer to the first reservoir boat facing the user in the two-column reservoir at A1,1.
86. Press enter to proceed. The system will now add 400 µL of RNA preparation buffer to rows and dispose of the used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in two transfers per row using the eight-channel 300 µL pipette.
87. Centrifuge the unsealed plate for 5 min at 3,000–5,000*g* at room temperature, taking care to ensure that the system is balanced. Discard the flow through and reassemble the filter and collection plate.
88. Add 84 mL of wash buffer to the second reservoir boat facing the user in the two-column reservoir at A1,1.
89. Press enter to proceed. The system will now add 700 µL of wash buffer to rows and dispose of the used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in three transfers per row using the eight-channel 300 µL pipette.
90. Centrifuge the unsealed plate for 5 min at 3,000–5,000*g* at room temperature, taking care to ensure that the system is balanced. Discard the flow through. Using a new collection plate, reassemble the filter and collection plate.
91. Add 48 mL of wash buffer to the second reservoir boat facing the user in the two-column reservoir at A1,1.
92. Press enter to proceed. The system will now add 400 µL of wash buffer to the rows and dispose of the used tips.
    - For the OT1, this will be done in one transfer per row using the eight-channel 1 mL pipette.
    - For the OT2, this will be done in two transfers per row using the eight-channel 300 µL pipette.
93. Centrifuge the unsealed plate for 5 min at a speed between 3,000 and 5,000*g* at room temperature, taking care to ensure that the system is balanced. Discard the flow through and dispose of the collection plate. Place the filter plate on a new 96-well PCR plate (the elution plate), taking care to align the wells such that the numbers match.
94. Dispense 12.5 µL of nuclease-free water to each sample well by hand using a P20 multichannel pipette and a disposable reagent reservoir, taking care to wet the filter only and not splash the sides.
95. Centrifuge the unsealed plate for 5 min at a speed between 3,000 and 5,000*g* at room temperature, taking care to ensure that the system is balanced.
96. During the centrifugation, label two 96-well PCR plates, each of which will be used for a separate preparation of the RNA, and two eight-well PCR strips, both of which will be used to confirm RNA quality on the Bioanalyzer. Place both 96-well PCR plates in cold plate holders.
97. After centrifugation, confirm that cfRNA has been eluted into wells as expected. Place the elution plate in a cold plate holder. Pipette 4 µL of eluted cfRNA into the corresponding well of each labeled plate. Seal the labeled plates and place in a −80 °C freezer. Pipette 1 µL of eluted cfRNA from randomly selected wells across plate (Step 1) into the eight-well PCR strips to be run on the Bioanalyzer. Cap the PCR strips and seal the original elution plate. Freeze everything at −80 °C.

Downstream processing

RNA quality check

● TIMING 1 h

98. If running clinical samples (Step 2, option B), select 11 wells at random across the plate, spanning both columns and rows, to be run on 1 Bioanalyzer chip. If running the RNA control (Step 2, option A), proceed to Step 101.

99. Prepare the samples and RNA Pico chip for the Bioanalyzer per the manufacturer's instructions. In our experience, we have found it easier to pipette 5 µL of RNA 6000 Pico marker into the eight-well PCR strip containing 1 µL of sample, mix it and then transfer a total of 6 µL into the corresponding chip well, as compared with pipetting the marker and sample sequentially. Avoid introducing bubbles in this process.
100. Assess sample quality following the Bioanalyzer run. Typically, we expect anywhere between 0.1 and 10 ng of total RNA extracted from 1 mL of plasma.
▲ **CRITICAL STEP** For exogenous RNA control samples, proceed to Steps 101–104. For frozen plasma samples, proceed to Steps 105–107.
◆ **TROUBLESHOOTING**

RT–qPCR using the exogenous RNA control

● **TIMING** 2 h

101. Prepare the master mix for the iTaq Universal Probes One-Step Kit following the manufacturer's instructions.
102. Combine 1 µL of isolated exogenous RNA (from Step 97) with 0.5 µL of the corresponding Taqman probes and 8.5 µL of master mix.
103. Perform RT–qPCR on the Bio-Rad CFX system (either 96 or 384) using the following program:

Cycle number	Program
1	50 °C, 10 min
2	95 °C, 1 min
3–43	95 °C, 10 s 60 °C, 20 s

104. Obtain the cycle thresholds from the corresponding CFX software and compare them with the expected results. This marks the end of the procedure when using an exogenous RNA control.
◆ **TROUBLESHOOTING**

RNA-sequencing library preparation using frozen plasma samples

● **TIMING** 2 d

105. Prepare cfRNA-sequencing libraries following manufacturer instructions for SMARTer Stranded Total RNAseq Kit v2 using 4 µL of eluted cfRNA from Step 97 as starting input.
◆ **TROUBLESHOOTING**
106. Barcode samples using unique dual index barcodes.
107. Pool in an equimolar fashion and sequence in paired-end mode (2 × 75 bp) to an average depth of 50 million reads per sample.

Troubleshooting

Troubleshooting advice can be found in Table 3.

Table 3 | Troubleshooting table

Step	Problem	Possible reason	Possible solution
13 (problem can occur at any subsequent step)	Robotic system remains uncalibrated after running through calibration steps	The system is not homed properly before calibration and/or the pipette is not secure	Recalibrate the system, taking care to home between different labware. Confirm that the pipette has no wiggle room.
	Pipette becomes miscalibrated in the middle of a run	Pipette is not securely fastened	If clinical samples, finish protocol by hand. After storing samples, confirm pipette calibration. Perform a dry run using Step 2A to confirm that the issue has been resolved.

Table 3 (continued) | Troubleshooting table

Step	Problem	Possible reason	Possible solution
27 (problem can occur at any subsequent step)	Robot tip pickup fails after accurately running for part of the procedure	Slight miscalibration of pipette position relative to tip location	Recalibrate ensuring that upon tip pickup for later rows (after row 6) you do not start to hear a slight slamming sound or see scuff marks on the side of the tips that have been picked up (both indicate that tip pickup is off by ~1 mm or less). It is crucial that tip pickup is smooth for all 12 rows.
50	Slurry runs through Norgen filter as opposed to remaining on top	Manufacturing error in filter	Continue to process sample. Make note of filter malfunction and call manufacturer.
100, 104	No RNA detected	Not enough EtOH in buffer, which led to RNA being eluted from the column before the final elution	Confirm EtOH concentration in buffer. If the problem remains unresolved, perform the procedure using the RNA control and confirm RNA concentration after first RNA concentration, DNAse digestion, and final elution step.
105	RNA detected but library preparation fails	Contaminants present in eluant inhibit reverse transcription	To diagnose the problem, combine 2 µL of the negative control sample (see Step 2B(iv)) from the problematic plate with 8 µL nuclease-free water. Add 500 ng control RNA (prepared in 'Reagent setup') to this diluted eluant sample. Perform RT–qPCR with probes for ERCC54. If cDNA is now detected by RT–qPCR after diluting your negative control sample, this confirms the presence of contaminants. To avoid introducing contaminants in the future, increase spin time when drying Zymo filter plate and ensure that the centrifuge reaches an appropriate speed of 3,000–5,000*g*.

Timing

Steps 1–2, Prepare samples: 10 min
Steps 3–64, RNA extraction: 3 h (1.5 h per 48-well plate)
Steps 65–68, DNA digestion: 20 min
Steps 69–97, RNA cleaning and concentration: 1 h
Steps 98–100, RNA quality check: 2 h
Steps 101–107, Downstream processing: 1–3 d

Anticipated results

This semi-automated protocol allows the user to extract cfRNA from 96 samples and confirm the sample quality in under 1 d. We previously used this protocol to extract cfRNA from 404 samples (Extended Data Fig. 1 in ref. 14). Key quality metrics include the number of reads that uniquely map to the human genome, number of reads assigned to genes, and two custom metrics of DNA contamination and RNA degradation calculated in our bioinformatic processing pipeline[25]. Extracting cfRNA and confirming sample quality from all samples required 27 h (one 96-well batch was processed over 2 d because of when samples were received). Preparing libraries for sequencing took another 10 d. Altogether, we prepared sequencing libraries for all 404 samples within 2.5 weeks as compared with the months that it would have required to process samples manually. We then processed this raw data to obtain counts tables. An example bioinformatic pipeline, which we used to process RNA-sequencing data[14], is provided in ref. 25. After obtaining a RNA-sequencing counts table, data analysis depends on the question at hand. Example analyses are included in our prior work and the associated repository[26] and include differential expression analysis and machine learning to predict risk of preeclampsia early in pregnancy.

Overall, we have shown that semi-automated sample processing can be performed affordably and quickly within individual academic laboratories. Shifting to semi-automated sample processing alleviates a major bottleneck in discovery-focused work and allows for seamless transitions from sample collection to data generation and analysis. With this robust and scalable solution, we can test hypotheses faster and spend less time on repetitive but important tasks such as sample processing.

Data availability

Source data are provided with this paper. This file includes raw data to reproduce Fig. 3b–d and Fig. 4. The Supplementary Manual also includes instructions for how to build a system similar to the OT1 from scratch.

Code availability

All software, including the data analysis required to generate Figs. 3 and 4, are available at https://github.com/miramou/cfRNA_pipeline_automation (https://doi.org/10.5821/zenodo.7931552) (ref. 24) and https://github.com/miramou/cfRNA_pipeline (https://doi.org/10.5281/zenodo.7931554) (ref. 25).

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Author contributions

M.N.M. conceptualized and designed this protocol, and collected and analyzed the data in collaboration with S.R.Q. All authors contributed to the writing and editing of the manuscript.

Competing interests

S.R.Q. is a founder, consultant and shareholder of Mirvie. M.N.M. is also a shareholder of Mirvie.

Additional information

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Key references using this protocol
Moufarrej, M. N. et al. *Nature* **602**, 689–694 (2022): https://doi.org/10.1038/s41586-022-04410-z
Video protocol
Moufarrej, M. (2020): https://youtu.be/g6RsSaNvSNA